strength. It is the business of the model to tell us whether such a fake signal will be evolutionarily stable, or whether natural selection will enforce decent, honest, and truthful advertising standards.

3. Unlike the females eyeing them, males do in a sense 'know' their own quality; and they adopt a 'strategy' for advertising, a rule for advertising conditionally in the light of their quality. As usual, by 'know' I don't mean cognitively know. But males are assumed to have genes that are switched on conditionally upon the male's own quality (and privileged access to this information is a not unreasonable assumption; a male's genes, after all, are immersed in his internal biochemistry and far better placed than female genes to respond to his quality). Different males adopt different rules. For instance, one male might follow the rule 'Display a tail whose size is proportional to my true quality'; another might follow the opposite rule. This gives natural selection a chance to adjust the rules by selecting among males that are genetically programmed to adopt different ones. The advertising level doesn't have to be directly proportional to the true quality; indeed a male could adopt an inverse rule. All that we require is that males should be programmed to adopt *some* kind of rule for 'looking at' their true quality and on the basis of this choosing an advertising level—size of tail, say, or of antlers. As to which of the possible rules will end up being evolutionarily stable, that again is something that the model aims to find out.

4. Females have a parallel freedom to evolve rules of their own. In their case the rules are about choosing males on the basis of the strength of the males' advertisement (remember that they, or rather their genes, lack the males' privileged view of the quality itself). For example, one female might adopt the rule: 'Believe the males totally.' Another female might adopt the rule: 'Ignore male advertising totally.' Yet another, the rule: 'Assume the opposite of what the advertisement says.'

So, we have the idea of males varying in their rules for relating quality to advertising level; and females varying in their rules for relating mate choice to advertising level. In both cases the rules vary continuously and under genetic influence. So far in our discussion, males can choose any rule relating quality to advertisement, and females can choose any rule relating male advertisement to what they choose. Out of this spectrum of possible male and female rules, what we seek is an evolutionarily stable pair of rules. This is a bit like the 'faithful/philanderer and coy/fast' model in that we are looking for an evolutionarily stable male rule and an evolutionarily stable female rule, stability meaning mutual stability, each rule being stable in the presence of itself and the other. If we can find such an evolutionarily stable pair of rules we can examine them to see what life would be like in a society consisting of males and females playing by these rules. Specifically, would it be a Zahavian-handicap world?

Grafen set himself the task of finding such a mutually stable pair of rules. If I were to undertake this task, I should probably slog through a laborious computer simulation. I'd put into the computer a range of males, varying in their rule for relating quality to advertisement. And I'd also put in a range of females, varying in their rule for choosing males on the basis of the males' advertising levels. I'd then let the males and females rush around inside the computer, bumping into one another, mating if the female choice criterion is met, passing on their male and female rules to their sons and daughters. And of course individuals would survive or fail to survive as a result of their inherited 'quality'. As the generations go by, the changing fortunes of each of the male rules and each of the female rules would appear as changes in frequencies in the population. At intervals I'd look inside the computer to see if some kind of stable mix was brewing.

That method would work in principle, but it raises difficulties in practice. Fortunately, mathematicians can get to the same conclusion as a simulation would by setting up a couple of equations and solving them. This is what Grafen did. I shall not reproduce his mathematical